

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.N o	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

/ VENNA MEENAKSHI REDDY (192471048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Venna Meenakshi Reddy

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding ; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Question 1:

Answer: (c) All attributes must be atomic

Explanation:

Consider a STUDENT table where one column stores multiple phone numbers such as 9876, 8765. This violates 1NF because the attribute contains multiple values. To achieve 1NF, each attribute must contain a single atomic value.

Question 2:

Answer: (b) Has no partial dependencies on primary key

Explanation:

Suppose (StudentID, CourseID) is a composite key and StudentName depends only on StudentID. This is a partial dependency. A relation in 2NF must first be in 1NF and must eliminate such partial dependencies.

Question 3:

Answer: (d) 4NF

Explanation:

If a student can have multiple skills and multiple hobbies, and these are independent multi-valued facts, a multi-valued dependency exists. 4NF is designed to eliminate non-trivial multi-valued dependencies.

Question 4:

Answer: (c) 3NF

Explanation:

Consider an FD $A \rightarrow B$ where A is not a superkey but B is a prime attribute. It may satisfy 3NF but violate BCNF. Therefore, BCNF imposes a stricter condition than 3NF: every determinant must be a superkey.

Question 5:

Answer: (b) X determines Y

Explanation:

If $\text{StudentID} \rightarrow \text{StudentName}$, then knowing the StudentID allows us to determine exactly one StudentName. Therefore, X determines Y in the functional dependency $X \rightarrow Y$.

Question 6:

Answer: (d) Retrieval anomaly

Explanation:

Suppose student and course information are unnecessarily repeated in one table. Changing a student's information can cause an update anomaly, removing a record can cause a deletion anomaly, and inability to insert independent information can cause an insertion anomaly. These are standard database anomalies; retrieval anomaly is not one of the standard normalization anomalies.

Question 7:

Answer: (b) Any set of attributes that uniquely identifies tuples

Explanation:

Suppose StudentID uniquely identifies a student. Then {StudentID} is a superkey. If {StudentID, StudentName} also uniquely identifies the student, it is also a superkey, although it is not minimal. Hence, a superkey can contain extra attributes.

Question 8:

Answer: (c) Transitively dependent on A

Explanation:

Given $A \rightarrow B$ and $B \rightarrow C$, A determines B and B determines C. Therefore, A indirectly determines C through B. This is called a transitive dependency:

$$A \rightarrow B \rightarrow C$$

Question 9:

Answer: (c) Project-Join Normal Form

Explanation:

Consider a relation that can be decomposed into several projections and reconstructed using joins without generating spurious tuples. 5NF, also called Project-Join Normal Form (PJ/NF), handles join dependencies that cannot be explained by lower normal forms.

Question 10:

Answer: (a) No data is lost during decomposition

Explanation:

Suppose STUDENT(StudentID, Name, Dept) is decomposed into STUDENT(StudentID, Name) and STUDENT_DEPT(StudentID, Dept). If joining the decomposed tables reconstructs the original information exactly, the decomposition is lossless.